**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 240 · Software testing: finding and correcting faults

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Define a cache miss.

Q2. When is A AND B true?

Q3. Give a boundary-invalid input just below 1.

Q4. Does 61 satisfy `seconds>=1 || seconds<=60`?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

logic error — Runs but produces the wrong result.

inclusive boundary — Endpoint allowed by a range.

validation — Check input meets constraints.

debugging — Find and correct a fault.

regression test — Repeat tests after a change.

## New learning

Read the stated rule — Because the permitted ages are inclusive, both age 12 and age 18 must be accepted.

Locate the Boolean error — The strict comparison operators > and < exclude both valid boundaries.

Repair and test — Change to >= and <=; retest the boundaries plus values just outside the range.

## Worked example

A score validator intended for 0–10 uses score>0 && score<10. Fix it.

1. Predict first — The valid endpoints were incorrectly rejected.

2. Model and check — Change the operators to >= and <=, and test −1 and 11 as invalid values.

# We Do · Practise with guidance

A duration validator for 1–60 accepts 61 because it uses OR. Correct it.

```
return seconds >= 1 && seconds <= 60;
```

Predict first

OR accepts 61 because only one side needs to be true.

Model and check

AND requires both limits to hold and correctly rejects 61.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Correct x>5 && x<15 for values 5–15 inclusive, and state whether 5 and 15 are valid.

### Question 2

Explain why `age>=12 || age<=18` accepts 100, correct it, and give outputs for 11,12,18,19.

### Question 3

Outline a debugging and regression-testing process after correcting a validator.

# Exit Ticket · Answer before checking

For a bounded range, OR usually replaces AND without changing the outcome.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Requested result is not already stored.

Q2. Only when both A and B are true.

Q3. 0.

Q4. Yes, because seconds>=1 is true.

### You Do pages · Independent questions

Q1. [4] Use the lower inclusive boundary x >= 5 [1] and upper inclusive boundary x <= 15 [1] together with AND: x >= 5 && x <= 15 ; x=5 is accepted [1] and x=15 is accepted [1]. Both clauses must be combined in the final condition.

Q2. [6] 100>=12 is true [1]; use && [1]; 11 false [1], 12 true [1], 18 true [1], 19 false [1].

Q3. [6] Reproduce failing case [1]; identify faulty condition [1]; correct code [1]; retest valid boundaries [1] and invalid neighbours [1]; retain automated regression tests [1].

### Exit Ticket · Model answer

False. OR often wrongly accepts values outside the range.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding